Github Link: https://github.com/marcus-cheema/CS-485/blob/main/Literature-Review.pdf
Marcus Cheema
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Banerjee, Nabaneeta. "IJIRT." *Synthesis of Arduino Based Musical Instrument*, IJIRT, July 2022, https://ijirt.org/publishedpaper/IJIRT155878_PAPER.pdf.

Published July 2022 in IJIRT, Nabaneeta Banerjee and others explore the integration of the Arduino technology to create a musical instrument. The study presents a practical approach to designing and building a musical instrument using the Arduino Uno, resistors, amplifiers, etc., while emphasizing its cost-effectiveness, versatility, and ease of customization. Their research details the hardware setup, including the use of sensors and actuators interfaced with Arduino, to generate musical notes and melodies. The researchers discuss the potential applications of such an instrument in educational settings and creative arts, offering insights into the technical and creative aspects of synthesizing music with Arduino technology.

Clary, Matt. "Interfacing to an LCD Screen Using an Arduino." *Interfacing to an LCD Screen Using an Arduino*, Michigan State University, 3 Apr. 2015, www.egr.msu.edu/classes/ece480/capstone/spring15/group05/uploads/4/7/5/1/47515639/ece_480_app_note_matt_clary.pdf.

Published April 2015 in Michigan State University, Matt Clary demonstrates how to connect an Arduino Uno microcontroller to an LCD display. It covers the pinout, wiring, and programming techniques needed to display text on the LCD. The study emphasizes the simplicity and versatility of this interface, allowing users to easily integrate LCDs into various projects. Matt also provides an introduction to the software libraries required for controlling the LCD display through an Arduino, further illustrating the LCD's usefulness for educational and DIY projects.

L, Yabesh Isak, and et al. "A Novel Toy Piano Using Arduino: EPRA International Journal of Multidisciplinary Research (IJMR)." *A NOVEL TOY PIANO USING ARDUINO | EPRA International Journal of Multidisciplinary Research (IJMR)*, EPRA International Journal of Multidisciplinary Research (IJMR), Nov. 2023, eprajournals.com/IJMR/article/11692/abstract.

Published November 2023, Yabesh Isak and others present a project that utilizes Arduino technology to create an interactive toy piano. It details the design, components, and programming required to build a musical instrument capable of producing various sounds. Moreover, the researchers provide detailed flowcharts for the execution/build steps. The study highlights the educational benefits of such projects, demonstrating how they can enhance learning in electronics and programming.

Madankar, Abhishek A., and et al. "Cost Effective Electronic Midi Pad Using Arduino Uno | IEEE Conference Publication | IEEE Xplore." *Cost Effective Electronic MIDI Pad Using Arduino UNO*, IEEE, 2023, ieeexplore.ieee.org/document/10169235/.

Published in 2023 in the Institute of Electrical and Electronics Engineers, Abhishek Madankar and others present a low-cost MIDI pad system built using Arduino UNO. The researchers bring awareness to issues of opportunity in the music industry, which can be attributed to the cost of instruments and instrument playing systems. It explores the design and implementation of a versatile electronic musical instrument that can be used for music production and live performances using the Arduino Mega 2560. The study emphasizes the affordability and simplicity of the project, offering insights into its practical applications in educational settings and DIY music projects.

Vieria, Rômulo. "(PDF) Fliperama: An Affordable Arduino Based MIDI Controller." *An Affordable Arduino Based MIDI Controller*, International Conference on New Interfaces for Musical Expression, Sept. 2020, www.researchgate.net/publication/344013716_Fliperama_An_affordable_Arduino_based_MIDI_Controller.

Published in September 2020 in the International Conference on New Interfaces for Musical Expression, Rômulo Vieria and Flávio Schiavoni explore Arduino technology to build a cost-effective MIDI controller. The project involves integrating an Arduino Pro Micro with a CD4067 multiplexer to expand output capabilities, along with various control interfaces like rotary potentiometers and Sanwa joystick buttons. These components allow the controller to replace traditional piano keys and drum pads in a cost-efficient manner. By eliminating the need for third-party software through direct USB-MIDI conversion, the project significantly reduces the overall cost, making it an accessible tool for musicians and students.